EF-Day03 Assignment

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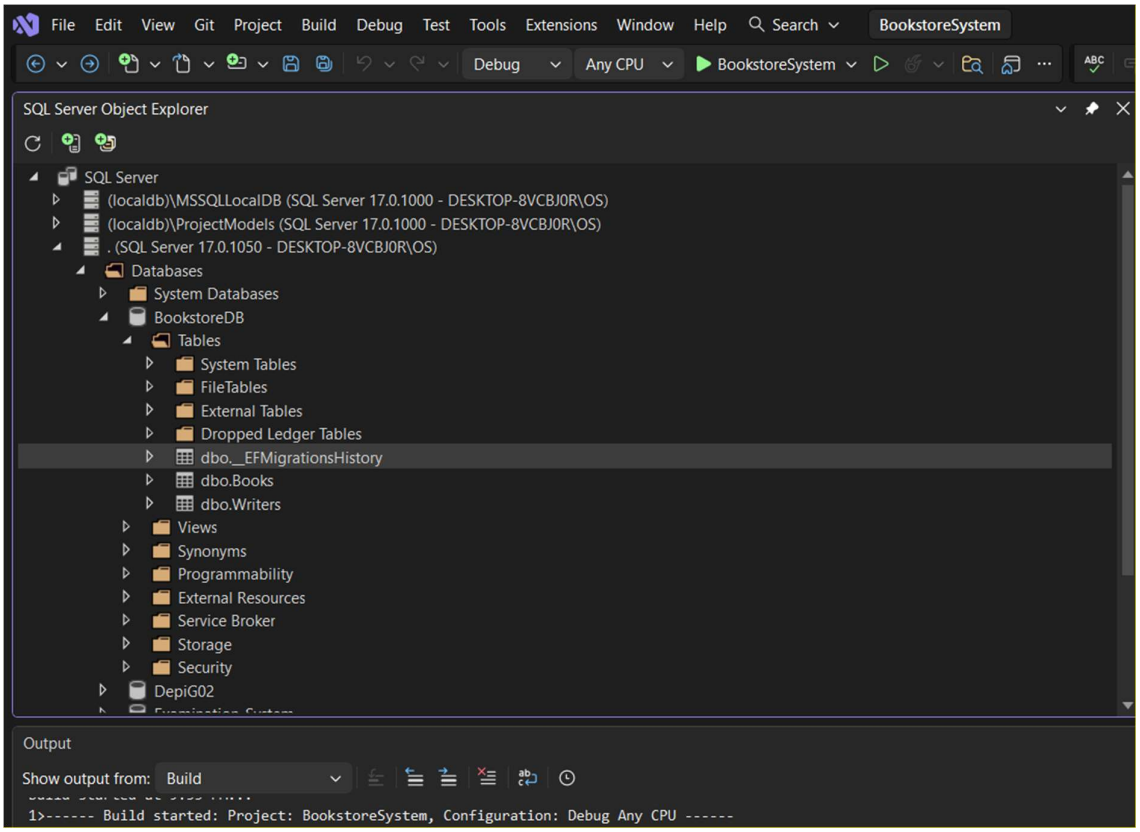
Screenshot of database and tables:-



EF01 Task — Theory Questions

Entity Framework Core | Code First Approach

Q1: Why Use Code First Rather Than DB First?

In **Code First**, we write C# classes and EF Core generates the database automatically. In **DB First**, the database is created manually and classes are generated from it. Code First is preferred because:

- **Version Control:** Migration files are tracked in Git with the code. Every schema change is recorded and reversible.
- **Productivity:** Developers work only in C#. Adding a column = adding a property, then running Add-Migration.
- **Migrations:** Schema changes are applied consistently across Dev, Staging, and Production.
- **Testability:** You can use InMemory provider for unit tests — no real SQL Server needed.
- **New Projects:** Ideal when starting from scratch with no existing database.

Q2: Why Use Fluent API Over Data Annotations?

Both configure EF Core mappings, but **Fluent API** is preferred in professional apps because:

- **Separation of Concerns:** Data Annotations mix database rules into domain classes. Fluent API keeps mapping in separate configuration classes, leaving models clean.
- **No Framework Dependency:** With Fluent API, model classes have zero EF Core imports — they are pure C# objects (POCOs).
- **More Options:** Fluent API supports things Data Annotations cannot: composite keys, value converters, query filters, and more.
- **Easier to Maintain:** One configuration class per entity is much easier to navigate in large projects than scattered attributes.
- **Testability:** Clean domain models with no attributes are simpler to unit test.